Fraction operations with models

Explore and build the idea

20 daily questions include C1 and C2. Use separate paper when helpful.

Lesson focus: I can use equal parts to explain fraction multiplication and two kinds of division.

R1. 7×8

R2. $96 \div 12$

R3. $\frac{1}{2} + \frac{1}{4}$

R4. $0.6 + 0.25$

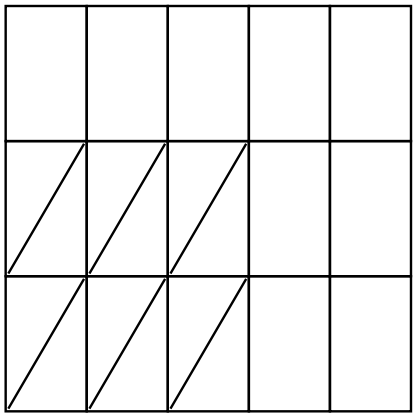
P1. Five bags each hold $\frac{3}{4}$ cup of seeds. How many cups altogether?

P2. A rectangle is $\frac{3}{5}$ meter long and $\frac{2}{3}$ meter wide. Find its area using equal parts.

P3. Find $\frac{2}{3}$ of $1\frac{1}{2}$ cups.

P4. Without multiplying first, compare $\frac{3}{4} \times 8$ and $\frac{5}{4} \times 8$ with 8. Then compute.

Lesson model



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Practice and compare methods

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P5. Three meters of ribbon are cut into $\frac{1}{4}$-meter pieces. How many pieces?

P6. A $\frac{1}{4}$-liter juice sample is shared equally among three cups. How much goes in each cup?

P7. Write one story for $2 \div \frac{1}{5}$ and one for $\frac{1}{5} \div 2$. Solve both.

F. A model has 4 columns and 3 rows. Shade $\frac{3}{4}$, then $\frac{2}{3}$ of it. Complete __ × __ = __.

D. Eli says $3 \div \frac{1}{4} = \frac{3}{4}$, because division makes numbers smaller. Show why this fails.

M1. 13×4

M2. $72 \div 9$

M3. $\frac{1}{3} + \frac{1}{6}$

Worked example - compare with the slide model

Find $\frac{3}{4}$ of $\frac{2}{3}$ cup.

Split a whole into 3 equal columns and 4 equal rows.
Select 2 columns, then 3 fourths of those columns.
The overlap is 6 of 12 pieces: $\frac{6}{12} = \frac{1}{2}$.

Result: $\frac{1}{2}$ cup

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Challenge, explain and check

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C1. Two batches each use $\frac{3}{4}$ liter of smoothie. All the smoothie is poured into $\frac{1}{4}$-liter cups. How many full cups? Explain with a model.

C2. A small rug is $1\ \frac{1}{2}$ meters by $\frac{2}{3}$ meter. A mat has area $\frac{4}{5}$ square meter. Which covers more floor, and by how much?

Finish the lesson

S. When I divide, I first ask whether I need the number of ___ or the size of ___.

E. A board is 5 meters long. Each shelf needs $\frac{1}{2}$ meter. How many shelves can be cut?